4. Crypt-Arithmetic Problem

Aim:

To write a Python program to solve the Crypt-Arithmetic problem by assigning digits to letters.

Algorithm:

1. Start the program.
2. Define the arithmetic equation.
3. Identify the unique letters.
4. Assign different digits to each letter.
5. Check that the first letter of each number is not zero.
6. Evaluate the arithmetic equation.
7. If the equation is correct, display the solution.
8. Stop.

Code:

```
from itertools import permutations

letters = "SENDMORY"

for p in permutations(range(10), 8):
    d = dict(zip(letters, p))

    if d["S"] == 0 or d["M"] == 0:
        continue

    send = d["S"]*1000 + d["E"]*100 + d["N"]*10 + d["D"]
    more = d["M"]*1000 + d["O"]*100 + d["R"]*10 + d["E"]
    money = d["M"]*10000 + d["O"]*1000 + d["N"]*100 + d["E"]*10 + d["Y"]

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Output:

```
==========
SEND = 9567
MORE = 1085
MONEY = 10652
```

Result:

Thus, the Python program to solve the Crypt-Arithmetic problem was executed successfully.